Iterative Decoding of Space-Time Differentially Coded Unitary Matrix Modulation

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***Abstract*—Noncoherent communication over the Rayleigh flat fading channel with multiple transmit and receive antennas is investigated. Codes achieving bit error rate (BER) lower than 10^{-4} at bit energy over the noise spectral density ratio (E_b/N_0) of 0.8 to 2.8 dB from the capacity limit were found with coding rates of 0.5 to 2.25 bits per channel use. The codes are serial concatenation of a turbo code and a unitary matrix differential modulation code. The receiver is based on a high-performance joint iterative decoding of the turbo code and the modulation code. Information-theoretic arguments are harnessed to form guidelines for code design and to evaluate performance of the iterative decoder.**

***Index Terms*—Capacity limit, differential modulation, joint iterative decoding, multiantenna, space-time, turbo-code, unitary matrix.**

I. Introduction

FADING channels are commonly used in wireless communications; see [1] and references therein. When line of sight is absent, the Rayleigh distribution is adopted in most cases to model the fading statistics. Recently, there has been considerable interest in the noncoherent multiple antenna communication channel, where fading coefficients that remain constant for the coherence interval T (in symbols) are unknown (to the transmitter and receiver) since the assumption that channel is known is questionable in the rapidly changing mobile environment. A general capacity achieving scheme is presented in [2] and [3], where it has been shown that for large signal-to-noise ratio (SNR), the capacity of multiantenna Rayleigh fading channel, where mutually independent equal energy fading coefficients are unknown (to the transmitter and receiver) is approached by unitary space-time block codes in which the signals transmitted by different antennas have equal energy and are mutually orthogonal in the coherence interval. In [4], a signaling scheme, for the Rayleigh flat-fading channel, of unitary space-time modulation is suggested for high SNR, or for $T \gg M$, as was also concluded in [2] and [3].

A modulation code for the multiple-input multiple-output (MIMO) channel was introduced in [5] and in [6], applying a general approach to differential modulation for the MIMO channel based on group codes. We design our demodulator motivated by the low-complexity receiver derived in [5] for the signaling scheme therein. The modulation scheme in [5] is shown to be equivalent to a scalar differentially encoded PSK [7] combined with a nonlinear space-time transmission rule [6], [8], [9], simplifying the encoding and decoding operations. We base our selection of constellation on the unitarity constraint as in [5]; however, we do not impose restrictions such as the group structure [5], [8] in an effort to approximate the capacity favored distribution, that is, the isotropical one [2], since the information rate is given higher priority over the spatial diversity gain in this work. See [10] for the tradeoff between rate and diversity over the coherent MIMO channel.

The concatenation of turbo code and a modulation code for the MIMO channel was introduced in [11]–[14]. It is evident that combining a Turbo code with a specific modulation code and incorporating efficient detection is not a trivial artifact of the known scalar channels or of the known coherent MIMO channels. In [11], an arbitrarily turbo coded modulation scheme (with a trivial constellation mapper as a modulation code) is presented, assuming known channel fading coefficients, and using a suboptimal decoding algorithm offering a significant improvement over the traditional space-time codes [15]. This is extended in [12], where a turbo code is concatenated with a unitary space-time modulator, assuming no channel information, and using a suboptimal receiver. We compare the performance of [12] to our suggested system in Section V.

The concatenated codes are decoded by the iterative algorithm designed originally for turbo codes in [16] and extended in later works [17]–[19] to the general case of interleaved concatenated codes. Further extensions of the iterative algorithm that place the decoding of additional elements specific to a coded communication system inside the iterative decoder were investigated in previous works, for example, [20] and [21] and references therein. The additional elements are treated similarly to another component code of a concatenated system, yielding decoder structures similar to ours. Some examples of such system specific elements are modulation codes and multiple symbol differential modulation over noncoherent channels [20], [21], and channel estimation [22]–[25]. In all the given examples, a significant improvement has been achieved over the separate detection of system elements and of the outer code.

Motivated by the results of [2] and using the concepts introduced in [6], we investigate a concatenated turbo encoder and modulation code for achieving reliable communication at SNR reasonably close to the capacity limit. The block fading channel models investigated are applicable for mobile wireless systems, TDMA systems, and for frequency-hopping systems. A differential modulation code related to [5] and [6] was used. It exploits the overlapping of observation intervals to greatly im-

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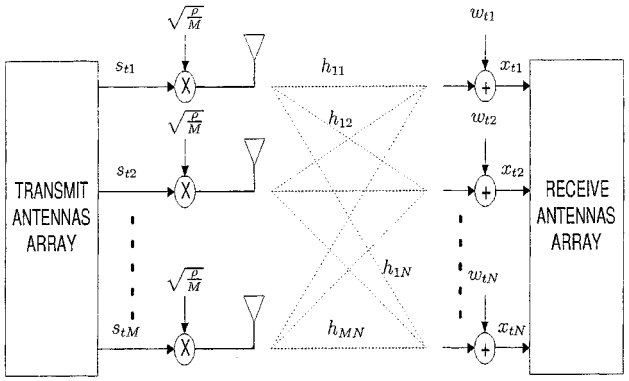


Fig. 1. Block diagram of a multiple antenna fading channel.

prove spectral and power efficiency over fading mobile channels. In scenarios where the overlapping cannot be used, such as in frequency-hopping systems, the same modulation code is still usable with reduced spectral and power efficiency.

In Section II, we present the system model, including the channel model, the transmitter structure, and the receiver. In Section III, we outline the guidelines for code selection and evaluation. In Section IV, we extend the observation block of the receiver for better utilization of time dependency. Section V includes simulation results over two channel models, and finally, conclusions in Section VI terminate the paper.

II. System Model

A. Channel Model

A channel model comprising M transmit and N receive antennas is illustrated in Fig. 1 and may be described by its baseband discrete time equivalent as

$$X = \sqrt{\frac{\rho}{M}} \cdot H \cdot S + W \tag{1}$$

where X is an $N \times T$ matrix of received signals, and S is an $M \times T$ matrix of transmitted signals, where s_{mt} is the signal transmitted from antenna m at time t. H is an $N \times M$ channel matrix. Its elements h_{nm} are independent complex Gaussian distributed with independent $CN(0,1)$ circularly symmetric complex Gaussian distributed elements. During a coherence block of T channel uses, the channel matrix H is fixed and unknown to the transmitter and the receiver. W is an $N \times T$ additive noise matrix, with independent $CN(0,1)$ circularly symmetric complex Gaussian distributed elements. Finally, ρ is the SNR at each receive antenna, whereas $E[|s_{mt}|^2] = 1$.

The investigated fading channel models follow.

1) *Non-overlap*: Successive coherence intervals are statistically independent.
2) *Overlap*: Fading is relatively slow. The independent fading coefficients are random complex Gaussian processes with a discrete auto-correlation function

$$R(k) = e^{-k \cdot T_s/\tau} \tag{2}$$

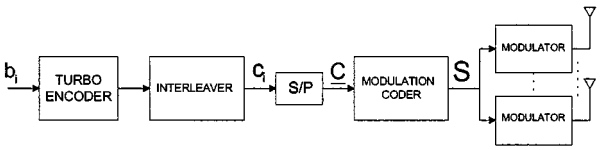


Fig. 2. Block diagram of a transmitter with turbo encoder and unitary matrix differential modulation coder.

where

- k discrete time index;
- T_s sampling period,
- τ fading time constant sufficiently large so that in every assumed coherence interval T, the variation of fading coefficients is small, approximating the fixed channel assumption.

In contrast with the *non-overlap* scheme, in this scheme correlation spans over several assumed coherence intervals.

In both scenarios $T = 2M$ is used. It is known from [3] that for $T/2 \geq \min\{M, N\}$ and large ρ, the capacity achieving S has orthonormal rows and a unitary isotropic distribution. From [2] and [3], it is clear that the capacity of the multiantenna channel can be approached for large T or ρ by code matrices with equal energy, isotropically distributed orthogonal rows. Accordingly, and as in [4], our attention was drawn to codes with the property [4]

$$S \cdot S^{\dagger} = T \cdot I \tag{3}$$

where S is an $M \times T$ *unitary code matrix*, and $S^{\dagger}$ stands for the complex conjugate transpose of S. The rows of S are kept as constraint free as possible to achieve distribution similar to the isotropical one; thus, we do not impose restrictions as, for example, in the group structure introduced in [5] and [6].

B. Transmitter

A block diagram of the transmitter, including a turbo encoder and a modulation encoder, is presented in Fig. 2. The input is a N_{inf} long stream of bits b_i. This bit stream is encoded by a standard turbo encoder, which includes a parallel concatenation of two convolutional encoders as in [16]. The turbo-encoded bits are bitwise interleaved and parsed into words $\underline{C}$ of length N_c bits. The interleaver adds essential time diversity in the *Overlap* case, disperses adjacent bit errors, and enables joint iterative decoding as described below. Each codeword $\underline{C}$ is mapped into an $M \times M$*unitary code matrix* D_i (or as later denoted $D(\underline{C})$) from the codebook of 2^{N_c} unitary matrices. The codebook in this work is any set of 2^{N_c} unitary matrices and is not restricted by a group structure [5]. An example of a *unitary code matrix* codebook for $M = 2$ is presented in Section V.

In the *non-overlap* case, a unitary matrix differential modulation is achieved by choosing an arbitrary unitary $M \times M$ reference matrix A_i. This setting is also known as the pilot symbol assisted modulation (PSAM) when the reference matrix A_i is the identity matrix. A transmission in each coherence interval consists of A_i as a reference followed by a differentially modulated matrix $A_{i-1} \cdot D_i$. It is clear that all possible $A_{i-1} \cdot D_i$ are unitary. Hence, the transmitted matrices S_i are given by

$$S_i = \begin{cases} \sqrt{M} \cdot A_i, & i = 1, 3, 5, \ldots, 2N_{\text{block}} - 1 \\ [S_{i-1} \cdot D_i], & i = 2, 4, \ldots, 2N_{\text{block}} \end{cases} \tag{4}$$

where the pair of matrices S_{i-1}, S_i is transmitted in a common coherence interval for each even i, and $N_{\text{block}} = \lceil N_{\text{inf}}/R_s T \rceil$ is the number of information matrices in a transmitted block originating from N_{inf} information bits at rate R_s bits per channel use and a coherence interval $T = 2M$. The term "channel use" includes the reference and information carrying symbols; thus, transmitting a matrix pair S_{i-1}, S_i from (4) involves T channel uses. It can be seen from [2] that the performance is invariant to the reference unitary matrix A_i used. A_i, D_i do not have to be square matrices [5]; however, nonsquare constellations are beyond the scope of this paper.

It is easy to show [6] that $[S_{i-1}S_i]$ conforms to the orthogonality property (3). D_i should approximate unitary uniform distribution, which is the optimal distribution according to [26]. Note that this differential structure does not use all the possible degrees of freedom inherent in $M \times 2M$ matrices [3] over a noncoherent channel. In the *overlap* case, the structure defined by (4) is retained, whereas A_i is the previous differential modulated unitary $M \times M$ matrix transmitted as in [6]. Hence, every $M \times M$ transmitted matrix is used twice: once as $[S_{i-1} \cdot D_i]$ in (4) and once as A_i in (4). Then, the transmitted $M \times M$ matrix for the *overlap* case is given by

$$S_i = \begin{cases} \sqrt{M} \cdot A_0, & i = 0 \\ [S_{i-1} \cdot D_i], & i = 1, \ldots, N_{\text{block}} \end{cases} \tag{5}$$

where $N_{\text{block}} = \lceil (N_{\text{inf}}/R_s T/2) \rceil$ is the number of information matrices in a transmitted block.

The *overlap* scheme clearly outperforms the *non-overlap* since it requires half of the transmit power to communicate at a double spectral efficiency. This approximation disregards the better time-diversity of the *non-overlap* scheme. The *non-overlap* scheme is also used here as a convenient tool for assessing performance of the receiver and comparing it with information-theoretic limits since in the *non-overlap* scheme, the channel and its capacity are well defined and may serve as a performance reference. The *non-overlap* scheme is suitable for scenarios in which the overlapping could not be used, such as in frequency-hopping systems; however, in this case, the structure of (4) may be released, and additional modulation codes may be used such as [12] or PSAM.

C. Receiver

1) General Structure: The block diagram of the receiver, including a demodulation decoder and a turbo decoder, is presented in Fig. 3, the notations of which are used in the following.

The receiver approximates the optimal joint detection of the turbo code and modulation code by means of iterative processing. Using principles presented, for example, in [20], the modulation decoder (MD) operating on the words $\underline{C}$ of the coded bits first computes soft metrics $ca_i = P(c_i = 1|X, cb_i)$ of the turbo encoded bits c_i, based on the two received matrices $X \triangleq [X_{i-1}, X_i]_{N \times T}$ and the extrinsic side information $cb_i = P(c_i = 1)$. The received matrices correspond to the two transmitted square matrices and are given by (1) and (4) $[X_{i-1}, X_i] = \sqrt{\rho/M} \cdot H \cdot [S_{i-1}, S_{i-1} \cdot D_i] + W$, where $i = 2, 4, \ldots, 2N_{\text{block}}$ in the *non-overlap* case and $i = 1, \ldots, N_{\text{block}}$ in the *overlap* case. The soft metrics ca_i are deinterleaved bitwise and decoded by one iteration of a turbo decoder [16], which also provides the soft metrics of the coded bits cb_i used by the modulation decoder in the next iteration. The turbo decoder is based on soft-in soft-out (SISO) modules as presented in [17] and also used in [20]. It includes two component code decoders, each decoding one of the component recursive convolutional codes by means of Bahl *et al.* algorithm [27]. This process is repeated on each received turbo-coded block of N_{inf} bits for N_{it} iterations.

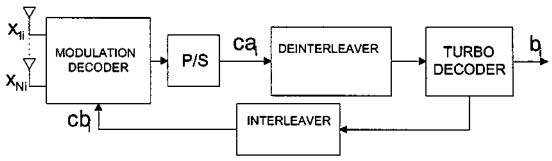


Fig. 3. Block diagram of iterative receiver.

2) Modulation Decoder: First, the joint probability $M(\underline{C}) \triangleq P(X, \underline{C})$ is computed for all possible codewords $\underline{C}$, based on the side information $cb_i = P(c_i = 1)$ and received $N \times T$ matrix $[X_{i-1}, X_i]$

$$M(\underline{C}) = P(X_{i-1}, X_i | D(\underline{C})) \cdot \prod_{j=1}^{N_c} P(c_j(\underline{C}) | cb_j) \tag{6}$$

where N_c is the bit-length of each codeword $\underline{C}$, and $P(c_i(\underline{C})|cb_i) = c_i \cdot cb_i + (1 - c_i) \cdot (1 - cb_i)$. The conditional pdf of the received matrices $[X_{i-1}, X_i]$ corresponding to the two square transmitted matrices in (4) can be derived similarly to [5, eq. (16)] as

$$P(X_{i-1}, X_i | \underline{C}) = \text{const} \cdot \exp\left\{ \frac{2\rho}{1 + \frac{T\rho}{M}} \cdot \text{real}\left\{ \text{tr}\left[X_{i-1} D(\underline{C}) X_i^{\dagger} \right] \right\} \right\} \tag{7}$$

where *const* contains irrelevant arguments of $P(X_{i-1}, X_i|\underline{C})$, which are identical for each codeword $\underline{C}$. The final bit metrics ca_i are calculated as in (10) and (11) of [20, eqs. (10) and (11)], marginalizing the probability of codewords $P(X, \underline{C})$ and extracting extrinsic information, while cancelling out the *const* term in (7); see also [20].

III. Code Evaluation Guidelines

Since the modulation encoder inputs are words $\underline{C}$ of turbo-coded and interleaved bits, all input words are practically equiprobable and statistically independent. Thus, to operate near capacity, a modulation code must achieve average mutual information (AMI) between the input words $\underline{C}$ and the channel output statistics X close to capacity with equiprobable independent input words $\underline{C}$. Thus, maximizing $I(\underline{C}; X)/T$, which is denoted as the word AMI, where $I(x; y) = E \log_2(P(x|y)/P(x))$ as in [28], happens to be an essential criterion in modulation code selection.

In the *non-overlap* scheme, the word AMI is also the code-restricted mutual information due to the strict independence between consecutive blocks. In the *overlap* scheme, the correlated fading channel, as introduced in (2), implies dependency between all the transmitted blocks, rendering capacity calculation rather complex; however, our receiver does not utilize this dependence beyond the observation interval T. Furthermore, the

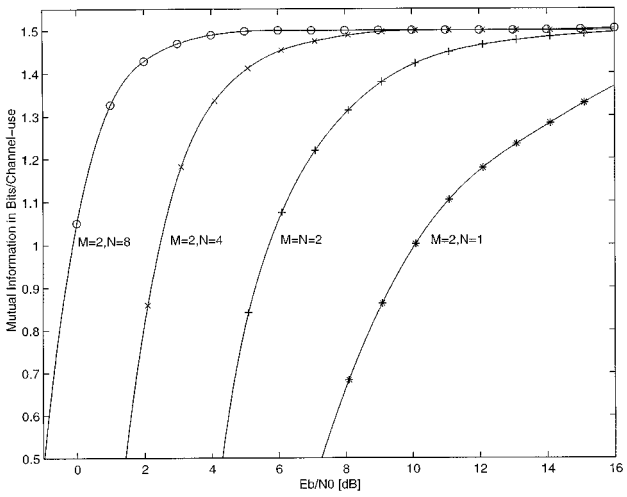


Fig. 4. Word AMI calculation in the *non-overlap* scheme for $T = 4$, two transmit antennas, and various number of receive antennas, using unitary code matrices and matrix differential modulation.

dependence is dispersed by the interleaver, and thus, for long blocks, the word AMI is equivalent to capacity as a performance bound, taking the limitations of the suboptimal receiver into account. The latter implies that full channel capacity is necessarily an upper bound for the word AMI.

In the simulations study throughout the paper, we use two transmit and two or four receive antennas, whereas the analysis is valid for any number of transmit and receive antennas and any modulation code. Fig. 4 demonstrates a calculation result for word AMI, in the *non-overlap* scheme, for $T = 4$, two transmit antennas and various number of receive antennas. The transmit matrices are constructed as in (4), where D_i is a 2×2 matrix containing QPSK elements. The mapping of 6-bit word $\underline{C}$ to unitary matrix D_i required in (4) is performed by mapping three pairs of bits to three QPSK symbols by Gray code, determining the fourth symbol by the unitary constraint. It is noticeable from Fig. 4 that for $M = 2$ transmit antennas, the greatest capacity improvement, by means of E_b/N_0, is achieved when multiplying the number of receive antennas from $N = 1$ to $N = 2$. When increasing the receive antenna diversity to $N = 4$ and $N = 8$, the capacity gain is still significant, even though the coherence interval is $T = 4$ and may be attributed to increased diversity.

The modulation decoder uses extrinsic information received from the turbo decoder to improve the input probability $ca_i = P(c_i = 1|X, cb_i)$ for the turbo decoder. At first iteration, the side information cb_i is missing; thus, the modulation decoder can at most produce $P(c_i = 1|X)$. The information content in $P(c_i = 1|X)$ is limited by $I(c_i; X)$, which is denoted in the following as bit-AMI and is clearly upper bounded by the word AMI. Hence, the initial conditions of the iterative decoder are limited by bit-AMI. The relevance of bit-AMI to the performance of an iterative decoder was also demonstrated in [20] and [29]. Thus, bit-AMI is an additional yet empiric criterion in modulation code selection. Clearly, performance with modulation feedback is upper bounded by word AMI and without modulation feedback by bit-AMI.

The word AMI was used to assess the expected performance of the modulation code to select the rate of the Turbo code and to assert that the decoder performs sufficiently close to the AMI bound. The bit-AMI led to selecting the Gray code mapping of symbols. The per-symbol Gray mapping is known to maximize the bit-AMI for a single antenna coherent communication [30] and achieved the best bit AMI for our MIMO case after examination of a few available mappings. It is also known [31] that minimum Euclidean distance between code-entailed $M \times T$ transmit matrices S_i (4) and (5) is tightly related with the error exponent, thus implying on further impact of symbol mapping selection.

IV. Extended Observation Interval

Two approaches are demonstrated for a better utilization of the channel characteristics for increasing the code-restricted mutual information. Both approaches achieve this goal by extending the observation interval at the receiver and by performing sub-optimal decoding. This method exploits the correlation in the fading random process and requires slower fading.

The first approach is denoted the reference differential encoding (RDE). In this method, a single reference matrix is used for differential encoding over a block of k transmitted matrices, i.e., $k - 1$ data carrying matrices are differentially modulated using a common arbitrarily chosen unitary reference matrix. A sub-optimal decoder, which keeps the decoding complexity constant for any k, is introduced hereby.

The second approach is based on the standard differential encoder (SDE), as presented in Section II-B. A sub-optimal decoder, which keeps the decoding complexity constant for any k, is suggested. The modulation decoder operates on k received matrices performing joint detection. The SDE approach does not require modification of the transmitter, and thus, the receiver may adjust the observation interval to the fading rate.

A. RDE-Based System

A unitary matrix differential modulation is achieved by choosing an arbitrary unitary $M \times M$ reference matrix A for each block. Every transmitted block matrix S consists of A as a reference followed by a differential modulated matrix $A \cdot D_1$, and $A \cdot D_2$, up-to $A \cdot D_{k-1}$. It is clear that all possible $A \cdot D_j$ are unitary. Hence, the $M \times kM$ transmitted matrix S is given by

$$S = \sqrt{M} \cdot [A \quad A \cdot D_1 \quad A \cdot D_2 \quad \ldots \quad A \cdot D_{k-1}] \qquad (8)$$

where D_j is the data carrying matrix, and A is the reference matrix. In each transmitted block, there are $N_{\text{block}} = \lceil ((k-1)N_{\text{inf}})/(kR_sM) \rceil$ information carrying matrices D_i in the *non-overlap* scheme and $N_{\text{block}} = \lceil N_{\text{inf}}/R_sM \rceil$ in the *overlap* scheme, where the overlapping extends over the first and last $M \times M$ transmitted matrices of adjacent blocks. Time indices are dropped intentionally for compactness of representation. The fading coefficients are assumed to be fixed during the transmission of S, which hereby defines a coherence interval of $T = k \cdot M$.

The received signal matrix may then be described as a block matrix

$$X = [\, X_0 \quad X_1 \quad \dots \quad X_{k-1} \,]. \tag{9}$$

The receiver has to compute $P(X|\underline{C})$ for all possible codewords $\underline{C}$ of length $(k-1)N_c$. This conditional pdf can be derived similarly to [5, eq. (16)] as

$$P(X|\underline{C}) = \text{const} \cdot \exp^{\{\frac{2\rho}{(1+T\rho)} \cdot \text{real}[\text{tr}(\text{XS}^\dagger \text{SX}^\dagger)]\}} \tag{10}$$

where *const* contains irrelevant arguments of $P(X|\underline{C})$, which are independent on S. It is clear from (10) that the main computation task is computing

$$\begin{aligned} \text{weight}(S) =& \text{real}\left\{\text{tr}\left[XS^\dagger SX^\dagger\right]\right\} \\ =& \text{real}\left\{\text{tr}\left[S^\dagger SX^\dagger X\right]\right\} \end{aligned} \tag{11}$$

for all possible $\underline{C}$. Rotating the matrix multiplication order in (11) is allowed because the trace is invariant to rotational reordering in multiplication of matrices. The matrix multiplication in (11) may be described in sub-matrix form, according to the definition of S in (8) and X in (9). Then, weight(S) is given by the real{tr[]} of

$$\begin{aligned} &\left[S^\dagger S\right] \cdot \left[X^\dagger X\right] \\ &= \begin{pmatrix} A^\dagger A & A^\dagger AD_1 & \dots & A^\dagger AD_{k-1} \\ D_1^\dagger A^\dagger A & D_1^\dagger A^\dagger AD_1 & \dots & D_1^\dagger A^\dagger AD_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ D_{k-1}^\dagger A^\dagger A & D_{k-1}^\dagger A^\dagger AD_1 & \dots & D_{k-1}^\dagger A^\dagger AD_{k-1} \end{pmatrix} \\ &\cdot \begin{pmatrix} X_0^\dagger X_0 & X_0^\dagger X_1 & \dots & X_0^\dagger X_{k-1} \\ X_1^\dagger X_0 & X_1^\dagger X_1 & \dots & X_1^\dagger X_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-1}^\dagger X_0 & X_{k-1}^\dagger X_1 & \dots & X_{k-1}^\dagger X_{k-1} \end{pmatrix}. \end{aligned} \tag{12}$$

The matrix multiplication result can be further simplified by using the unitarity property in (3) and by dropping the additive elements dependent only on X and not on D. Then, weight(S) is given by

$$\begin{aligned} \text{weight}(S) =& \text{const} + \\ & 2\text{real}\left\{\text{tr}\left[\sum_{i=1}^{k-1} D_i X_i^\dagger X_0 \right.\right. \\ & + \sum_{i=2}^{k-1} D_1^\dagger D_i X_i^\dagger X_1 \\ & + \sum_{i=3}^{k-1} D_2^\dagger D_i X_i^\dagger X_2 + \cdots \\ & \left.\left. + D_{k-2}^\dagger D_{k-1} X_{k-1}^\dagger X_{k-2}\right]\right\} \end{aligned} \tag{13}$$

where *const* represents the X dependent elements. Since the receiver has to compute weight(S) for all $2^{(k-1)N_c}$ possible codewords, the reduction of the decoding complexity is essential.

In order to turn the decoding process into a realistic task, it is suggested that we compute weight(S) for each data carrying matrix D_i $(0 \le i \le k-1)$ separately, using the estimation of all the other $\{D_j\}_{j=0;j\neq i}^{j=k-1}$ instead of their real value. The expected value $\{E[D_j]\}_{j=0;j\neq i}^{j=k-1}$ can be directly calculated from the extrinsic information received from the turbo decoder at each iteration, as shown later in (16).

Thus, the expression of weight(S) may be approximated by $\text{weight}_{\text{est}}(D_i)$ when using the estimations of some of codewords and further simplified by dropping the irrelevant elements for decoding, which are the additive elements dependent only on X or $E[D_j]$. Hence, for the information matrix D_{k-1}

$$\begin{aligned} \text{weight}_{\text{est}}(D_{k-1}) =& \text{const}_1 + \\ & 2\text{real}\left\{\text{tr}\left[D_{k-1}X_{k-1}^\dagger X_0 + E\left[D_1^\dagger\right]\right.\right. \\ & \times D_{k-1}X_{k-1}^\dagger X_1 + \cdots \\ & \left.\left. + E\left[D_{k-2}^\dagger\right] D_{k-1}X_{k-1}^\dagger X_{k-2}\right]\right\} \\ & \Downarrow \\ \text{weight}_{\text{est}}(D_{k-1}) =& \text{const}_2 + \\ & 2\text{real}\left\{\text{tr}\left[D_{k-1}X_{k-1}^\dagger \right.\right. \\ & \left.\left. \cdot \left(X_0 + \sum_{i=1}^{k-2} X_i E\left[D_i^\dagger\right]\right)\right]\right\} \end{aligned} \tag{14}$$

where const_1 includes elements dependent on X only, and const_2 also includes elements dependent on the combination of X and $E[D_j]$ for $j \neq k-1$. Equation (14) represents the sub-optimal expression for *weight* required for computing the probability of the codeword $\underline{C}$, which depends only on D_{k-1} $(D_{k-1}(\underline{C}))$. The process is repeated for each $\{D_j\}_{j=0}^{j=k-1}$ and thus allows computation of approximated probabilities for codewords of length N_c instead of $k \cdot N_c$. The generalized expression of $\text{weight}_{\text{est}}(D_j)$ for $(1 \le j \le k-1)$ is given by

$$\begin{aligned} &\text{weight}_{\text{est}}(D_j) = \text{const}_2 + \\ &2\text{real}\left\{\text{tr}\left[D_j X_j^\dagger \cdot \left(X_0 + \sum_{i=1, i\neq j}^{k-1} X_i E\left[D_i^\dagger\right]\right)\right]\right\}. \end{aligned} \tag{15}$$

The estimation of D_i is given by the optimal mean square error (MSE) estimator. Given the codeword probabilities computed from the turbo decoder extrinsic information [see also (6)], the estimated data matrices are

$$E[D_i] = \sum_{\underline{C}} P(\underline{C}|\underline{CB}) D(\underline{C}) \tag{16}$$

where $P(\underline{C}|\underline{CB})$ is calculated from the vector of the coded bits probabilities $\underline{CB}$, received from the turbo decoder. Then, expression $X_0 + \sum_{i=1, i\neq j}^{k-1} X_i E\left[D_i^\dagger\right]$ in (15) replaces X_0 as an improved reference matrix for computing the conditional probability of the two received $N \times M$ matrices given a codeword $\underline{C}$, as in (7). Clearly, the *non-overlap* case (4) and the corresponding receiver (7) are identical to the presented RDE system (8) and (15), when the observation interval is $T = 2M$.

A further modification that was found to be beneficial for improved simulation performance is to give different weighting for the estimated reference than the received reference in (15). When $\text{weight}_{\text{est}}(D_j)$ is given by

$$\text{weight}_{\alpha_{\text{est}}}(D_j) = \text{const}_2 + 2\text{real}\left\{\text{tr}\left[D_j X_j^{\dagger}\cdot\left(X_0+\alpha\cdot\sum_{i=1,i\neq j}^{k-1} X_i E\left[D_i^{\dagger}\right]\right)\right]\right\} \quad (17)$$

where α is the empirical weighting factor of the estimated reference. The best simulation performance was achieved for $\alpha = 1/(k-1)$, which would reduce the contribution of the estimated reference term to that of X_0 in the case of maximal $E\left[D_i^{\dagger}\right]$.

B. SDE-Based System

This system uses standard differential encoding for matrix differential modulation and uses an extended observation interval in the receiver. It can be used in the *overlap* and *non-overlap* cases. A reduced complexity sub-optimal decoder is derived here.

The sub-block of successive k transmit matrices, in the *non-overlap* case, is given by

$$S = \sqrt{M}\cdot\left[A \quad A\cdot D_1 \quad A\cdot D_1\cdot D_2 \quad \dots \quad A\cdot\prod_{j=1}^{k-1} D_j\right] \quad (18)$$

where D_j is the data carrying matrix, and A_i is the reference matrix. All D_j and A are $M\times M$ unitary square matrices. Equation (18) is equivalent to (5) in the *overlap* scheme. Its main difference from (4) is that instead of a single data carrying matrix (D_j) after the reference matrix $k-1$, data matrices are transmitted in the *non-overlap* scheme. The receiver assumes that the fading coefficients remain fixed during the transmission of S, as in Section IV-A.

Let us define the received block matrix for deriving the reduced-complexity SDE receiver by

$$X = \left[X_0 \quad X_1 \quad \dots \quad X_{k-1}\right] \quad (19)$$

where each element in the received matrix is an $N\times M$ received matrix. We need to compute $\text{weight}(S)$ as specified by (11). When multiplying $S^{\dagger}S$ and $X^{\dagger}X$ and omitting the discrete time index i for compactness of the representation, $\text{weight}(S)$ is given by the $\text{real}\{\text{tr}[]\}$ of (20), shown at the bottom of the page. The matrix multiplication result can be further simplified by using the unitarity property (3); thus, (20) is given by

$$\begin{aligned}&\left[S^{\dagger}S\right]\cdot\left[X^{\dagger}X\right]\\ &= M\cdot\begin{pmatrix} I & D_1 & \dots & \prod_{j=1}^{j=k-1} D_j\\ D_1^{\dagger} & I & \dots & \prod_{j=2}^{j=k-1} D_j\\ \vdots & \vdots & \ddots & \vdots\\ \prod_{j=k-1}^{j=1} D_j^{\dagger} & \prod_{j=k-1}^{j=2} D_j^{\dagger} & \dots & I\end{pmatrix}\\ &\quad\cdot\begin{pmatrix} X_0^{\dagger}X_0 & X_0^{\dagger}X_1 & \dots & X_0^{\dagger}X_{k-1}\\ X_1^{\dagger}X_0 & X_1^{\dagger}X_1 & \dots & X_1^{\dagger}X_{k-1}\\ \vdots & \vdots & \ddots & \vdots\\ X_{k-1}^{\dagger}X_0 & X_{k-1}^{\dagger}X_1 & \dots & X_{k-1}^{\dagger}X_{k-1}\end{pmatrix}. \end{aligned} \quad (21)$$

The matrix multiplication result can be further simplified by dropping the additive elements dependent only on X_i. Then, $\text{weight}(S)$ is given by

$$\begin{aligned}\text{weight}(S) =& \text{const} + 2\text{real}\left\{\text{tr}\left[\sum_{i=1}^{k-1}\prod_{j=1}^{j=i} D_j X_i^{\dagger}X_0 + \sum_{i=2}^{k-1}\prod_{j=2}^{j=i} D_j X_i^{\dagger}X_1 + \cdots + D_{k-1}X_{k-1}^{\dagger}X_{k-2}\right]\right\}\\ =& \text{const} + 2\text{real}\left\{\text{tr}\left[\sum_{m=0}^{k-2}\sum_{n=1}^{k-1}\prod_{j=m+1}^{n} D_j X_n^{\dagger}X_m\right]\right\}. \end{aligned} \quad (22)$$

In order to turn the decoding process into a realistic task, it is suggested to compute $\text{weight}(S)$, similarly to RDE (Section IV-B), for each data carrying matrix D_i $(0\le i\le k-1)$, using the estimation of all the other $\{D_j\}_{j=0:j\neq i}^{j=k-1}$ instead of their real value. The expected value $\{E[D_j]\}_{j=0:j\neq i}^{j=k-1}$ can be directly calculated from the extrinsic information received from the turbo decoder at each iteration (16). Thus, the expression of $\text{weight}(S)$ may be approximated by $\text{weight}_{\text{est}}(D_i)$ by using the estimations of some of codewords and further simplified by dropping the irrelevant elements for decoding from (22).

$$\begin{aligned}\left[S^{\dagger}S\right]\cdot\left[X^{\dagger}X\right] =& \begin{pmatrix} S_0^{\dagger}S_0 & S_0^{\dagger}S_0D_1 & \dots & S_0^{\dagger}S_0\prod_{j=1}^{j=k-1} D_j\\ D_1^{\dagger}S_0^{\dagger}S_0 & D_1^{\dagger}S_0^{\dagger}S_0D_1 & \dots & D_1^{\dagger}S_0^{\dagger}S_0\prod_{j=1}^{j=k-1} D_j\\ \vdots & \vdots & \ddots & \vdots\\ \prod_{j=k-1}^{j=1} D_j^{\dagger}S_0^{\dagger}S_0 & \dots & \dots & \prod_{j=k-1}^{j=1} D_j^{\dagger}S_0^{\dagger}S_0\prod_{j=1}^{j=k-1} D_j\end{pmatrix}\\ &\cdot\begin{pmatrix} X_0^{\dagger}X_0 & X_0^{\dagger}X_1 & \dots & X_0^{\dagger}X_{k-1}\\ X_1^{\dagger}X_0 & X_1^{\dagger}X_1 & \dots & X_1^{\dagger}X_{k-1}\\ \vdots & \vdots & \ddots & \vdots\\ X_{k-1}^{\dagger}X_0 & X_{k-1}^{\dagger}X_1 & \dots & X_{k-1}^{\dagger}X_{k-1}\end{pmatrix}. \end{aligned} \quad (20)$$

Decoding D_i then implies the calculation of the following *weight*

$$\text{weight}_{\text{est}}(D_i) = \text{const}_1 + 2\text{real}\left\{\text{tr}\left[\sum_{m=0}^{i-1}\sum_{n=i}^{k-1}\prod_{m+1\le j_1\le n} E[D_{j_1}]\cdot D_i \cdot \prod_{i+1\le j_2\le n} E[D_{j_2}]X_n^{\dagger}X_m\right]\right\} \quad (23)$$

and when $m+1>n$, the term $\prod_{j=m+1}^{j=n} E[D_j]$ is replaced by I. In the *overlap* case, $\text{weight}_{\text{est}}(D_i)$ can be calculated for the same i, i.e., the k-observation block may be shifted by one (a sliding window) for every D_i, thus eliminating the boundary effect of computing D_{k-1} using estimations of previous received matrices *only*.

An example for the *overlap* case, with $k=4$ and $i=2$, is chosen to demonstrate (23), which takes the form

$$\text{weight}_{(\text{est})}(D_2) = 2\text{real}\left\{\text{tr}\left[\left(D_2X_2^{\dagger}X_1 + E[D_1]D_2X_2^{\dagger}X_0 + D_2E[D_3]X_3^{\dagger}X_1 + E[D_1]D_2E[D_3]X_3^{\dagger}X_0\right)\right]\right\} \quad (24)$$

and includes only four elements for summation.

Clearly, the decoders given by (22) and (23) are identical to (7) when the observation interval is $T=2M$.

A further modification was found beneficial for improved simulation performance, like in the RDE, that is to give different weighting for the estimated product $X_1^{\dagger}X_2$, than the received product in (24). This results in

$$\text{weight}_{(\alpha_{\text{est}})}(D_2) = 2\text{real}\left\{\text{tr}\left[D_2X_2^{\dagger}X_1 + \alpha\left(E[D_1]D_2X_2^{\dagger}X_0 + D_2E[D_3]X_3^{\dagger}X_1\right) + \beta\left(E[D_1]D_2E[D_3]X_3^{\dagger}X_0\right)\right]\right\} \quad (25)$$

where α is the empirical weighting factor. The best simulation performance was achieved for $\alpha = 1/(k-1)$, $\beta=0$ in all simulations, which suggests that no improvement is expected by further increasing the observation interval, except for the use of one reference matrix for more information symbols in the *non-overlap* case only.

V. Simulation Performance

We present the performance of the turbo-coded unitary matrix differential modulation for two transmit and two or four receive antennas, compared with the capacity limit in Figs. 6–8. The bit error rate (BER) is measured for the bit energy over the noise spectral density E_b/N_0, which is defined as

$$\frac{E_b}{N_0} \triangleq \frac{\text{SNR}}{R_s} \equiv \frac{\rho}{R_s}.$$

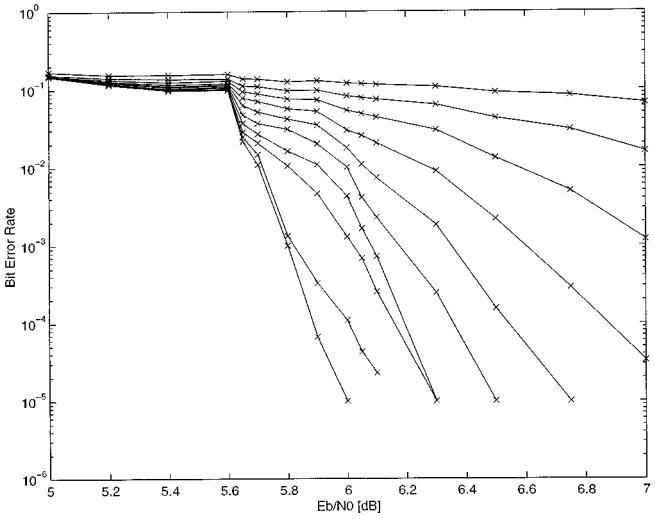


Fig. 5. Performance of the iterative receiver with $N_{\text{it}}=10$ iterations, $R_s=0.75$, *non-overlap*, $N_{\text{inf}}=3000$, $M=N=2$.

where R_s is the system rate given in uncoded information bits per channel use.

The component codes of the turbo code are recursive systematic convolutional codes, which are described by $(g_0,g_1)=(5,7)_{\text{octal}}$, where g_0 and g_1 are feedforward and feedback generating polynomials. The turbo-code employs a uniform pseudo random interleaver [18]. The performance was evaluated for three different turbo-code rates $R_c=(1/3),(1/2),(3/4)$. The interleaver of the turbo-coded bits is also a uniform random interleaver over the transmitted block, and both interleavers are selected randomly for each simulation block. The block size is $N_{\text{inf}}=3000$. For $R_c=(1/2)$, a block of $N_{\text{inf}}=15\ 000$ was simulated as well. The iterative decoding included $N_{\text{it}}=10$ iterations for all simulated results.

Two transmit antennas, along with two or four receive antennas, are used, with QPSK symbols at each transmit antenna. The mapping of 6-bit word $\underline{C}$ to unitary matrix D_i required in (4) is performed by mapping three pairs of bits to three QPSK symbols by Gray code. The 3-QPSK symbols are three elements of the 2×2 matrices D_i, and the fourth symbol is determined by the unitarity constraint. Thus, the modulation rate is 1.5 and three coded bits per channel sample for the *non-overlap* and the *overlap* scheme, respectively. The system rates corresponding to the turbo-code rates are $R_{s(\text{NoOverlap})}=(1/2),(3/4),(9/8)$ [bits/channel use] for the *non-overlap* scheme and $R_{s(\text{Overlap})}=1,(3/2),(9/4)$ [bits/channel use] for the *overlap* scheme. The per-symbol-Gray mapping described above is known to maximize the bit-AMI for a single antenna coherent communication [30] and achieved the best bit AMI for our MIMO case after examination of a few available mappings.

Fig. 5 demonstrates a typical performance of the iterative receiver in the *non-overlap* scheme, with $R_c=(1/2)$, $R_s=0.75$, $N_{\text{inf}}=3000$ with a modulation feedback. The performance resembles qualitatively that of a standard turbo decoder [16], improving over iterations and exhibiting a steep BER versus E_b/N_0 slope. Simulations in this figure are limited by a BER

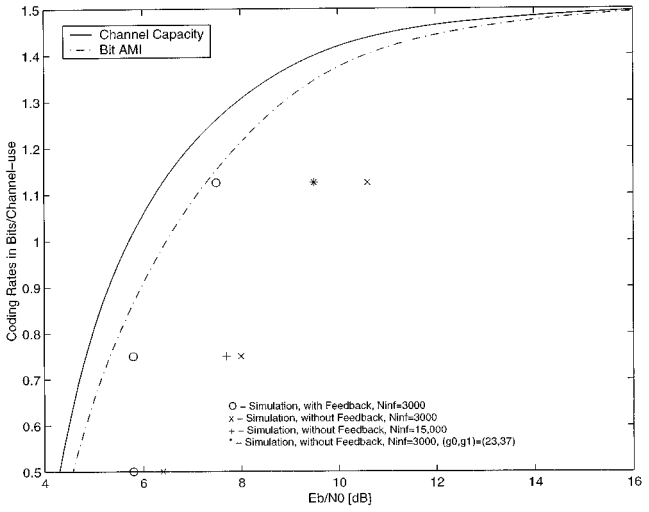


Fig. 6. *Non-overlap* scheme. Simulation results for BER $= 10^{-4}$, compared with word AMI, and bitwise-AMI bounds $N_{\rm it} = 10$, $M = N = 2$, $R_s = 0.5, 0.75, 1.125$.

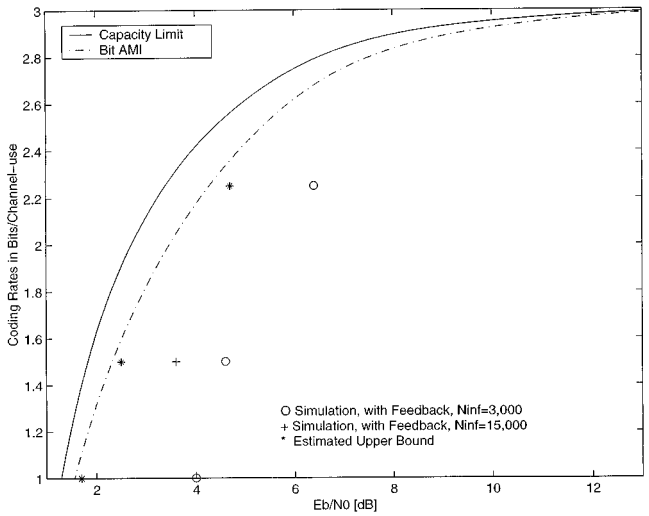


Fig. 7. *Overlap* scheme. Simulation results for BER $= 10^{-4}$, compared with word AMI and bitwise-AMI. $\tau = 64$, $N_{\rm it} = 10$, $M = N = 2$, $R_s = 1$, 1.5 and 2.25.

of 10^{-5}. An error floor was not evident in the range of BER investigated; however, it may appear at lower BER values. It can be seen that the system operating point at which BER is less than 10^{-5} is only 0.8 dB from the code-constrained channel capacity (see Fig. 6).

A. Non-Overlap Scheme

In Fig. 6, we present the performance of the described system in the *non-overlap* scheme, demonstrating the efficiency of the turbo code and the sub-optimal joint iterative decoder. It can be seen that the system operating points at which BER is less than 10^{-4} are only 0.8–1.4 dB from the code-constrained channel capacity. The upper bound on the simulated system performance is determined by the word AMI when exploiting the extrinsic information of the turbo-decoder, i.e., when using the modulation feedback (see Section III) and by the bit AMI bound in a system without modulation feedback.

The contribution of the iterative joint demodulator and turbo decoder is larger than predicted by the gap between the bit AMI and the word AMI, as clearly seen from Fig. 6, where operating points at which BER is less than 10^{-4} are 1.7–3.2 dB from bit AMI bound without modulation feedback. It is, however, evident that using a stronger turbo-code $(g_0, g_1) = (23, 37)_{\rm octal}$, when modulation feedback is not used, improves performance dramatically. In particular, for the higher rate turbo-code, an improvement of 1.1 dB is observed. This result suggests also that the concatenation of a trivial turbo-code with a modulation code creates a stronger code. A joint iterative receiver then performs near the code-restricted capacity limit, whereas a receiver without the modulation feedback fails to efficiently decode the concatenated codes, and thus, a stronger turbo-code is required.

Increasing the block length of the data from $N_{\rm inf} = 3000$ to $N_{\rm inf} = 15\,000$ yield a 0.3-dB improvement, suggesting that for longer blocks, the performance may get close to bit AMI bound. A similar phenomenon is evident in [32], where the minimum required SNR resulting from the random coding exponent as

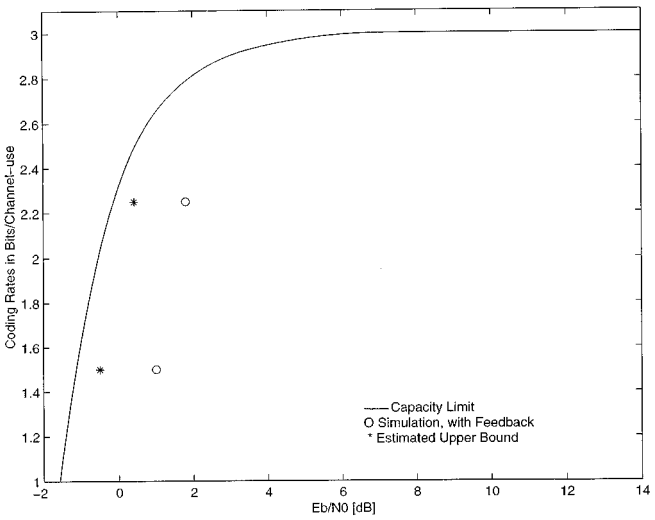


Fig. 8. *Overlap* scheme. Simulation results for BER $= 10^{-4}$, compared with word AMI, $N_{\rm inf} = 3000$, $\tau = 64$, $N_{\rm it} = 10$, $M = 2$, $N = 4$, $R_s = 1.5$ and 2.25.

defined by Gallager [28], which in turn leads to a word error rate lower than a given threshold, is computed. It is shown by means of error exponent that the SNR required to achieve a given error rate increases with decreasing block-length faster for bitwise decoding (see [32, Fig. 1, BICM]) than for wordwise decoding (see [32, Fig. 1, MLD]). This observation is in agreement with our results since the iterative receiver attempts to approximate optimal decoding.

B. Overlap Scheme

In Figs. 7 and 8, the performance of the system in the *overlap* scheme is evaluated. It should be noted that the channel capacity is calculated for a hypothetical channel, such that the fading coefficients are fixed for two consecutive matrix transmissions and

TABLE I
SIMULATION RESULTS SUMMARY OF RDE, SDE SCHEMES VERSUS THE *OVERLAP AND NON-OVERLAP* SCHEMES $N_{\rm it} = 10$, $N_{\rm inf} = 3000$, $M = N = 2$

Channel Model	System Model	T [Channel Uses]	$E_b/N_0[dB]$ $for BER \leq 10^{-4}$	R_s [Bits per Channel Use]
Non	RDE/SDE, $k = 2$	$2M$	5.8	0.5
Overlap	RDE, $k = 5$	$5M$	3.2	0.8
	SDE, $k = 5$	$4M$	3.4	0.8
Overlap	SDE, $k = 2$	$2M$	4	1
$\tau = 64$	SDE, $k = 4$	$4M$	3.4	1

independent on any previous transmission, yielding an overestimation of capacity. However, the channel was characterized more realistically for simulation by the correlation function (2).

A first-order estimation of the SNR loss, due to the variance of the fading coefficient in a single transmitted block, was done. For clarity, we analyze the one-dimensional case, and the same results can be derived for the multidimensional case. It can be shown that the channel-correlation function (2) determines the discrete random process of the fading as

$$H_i = (1-\alpha)\cdot H_{i-1} + \sqrt{\alpha\cdot(2-\alpha)}\cdot W_i \qquad (26)$$

where $\alpha = 1 - e^{-T_s/\tau}$ [thus, $R(k) = (1-\alpha)^k$ as in (2)], and W_i is the additive complex Gaussian noise with independent $CN(0,1)$ distribution. Thus, the received signal in (1) may be represented as

$$X_i = (1-\alpha)\cdot\sqrt{\frac{\rho}{M}}\cdot H_{i-1}\cdot S_i + \sqrt{\alpha\cdot(2-\alpha)\cdot\frac{\rho}{M}}\cdot S_i\cdot W_{i,1} + W_{i,2} \qquad (27)$$

where S_i is the single-dimensional transmit symbol at time i. H_i is given by (26). $W_{i,1}$ and $W_{i,2}$ are the additive complex Gaussian noise with independent $CN(0,1)$ distribution. This representation resembles the AR model representation (10) in [33]. Clearly, the first term in (27) is the main attenuated signal, and the second term is additional uncorrelated noise.

The SNR loss $L_{\rm Overlap}$ may now be computed by

$$L_{\rm Overlap} = \frac{1+\alpha\cdot(2-\alpha)\cdot\rho}{(1-\alpha)^2}. \qquad (28)$$

Result (28) is denoted in Fig. 7 as an estimated upper bound, which requires SNR higher by 0.45–1.3 dB than the word AMI bound for the different rates and SNRs. Some of the additional degradation could be caused by the correlation of the fading coefficients between observation intervals, reducing the time diversity of the scheme. Therefore, it is interesting to see the difference in performance between the case of $N = 2$ (Fig. 7) and $N = 4$ (Fig. 8) for $N_{\rm inf} = 3000$. For the latter ($N = 4$), the performance is 1.4–1.5 dB from the estimated upper bound, in contrast to 2.5 dB for $N = 2$, which indicates that the time diversity was replaced by additional space diversity. It is also interesting to notice that increasing the block length of the data from $N_{\rm inf} = 3000$ to $N_{\rm inf} = 15\ 000$, in the case of $N = 2$, yields a 1-dB better performance and only a 1.1 dB gap from the estimated upper bound at BER less than 10^{-4}.

C. SDE and RDE Schemes

In simulations of RDE and SDE systems in this section, the turbo-code rate chosen is $R_c = (1/3)$, which is expected to be most robust in means of error correction; hence, it conveys maximum side information for the modulation decoder. In Table I, the performance of the increased observation interval *non-overlap* schemes RDE $k = 5$ and SDE $k = 5$ are compared to that of the basic *non-overlap* scheme, as described in Section II and denoted in the table as SDE/RDE, $k = 2$. In the RDE, the $k = 5$ system, which is a common $M \times M$ reference matrix, was used to modulate four consecutive $M \times M$ matrices. The SDE $k = 5$ system also has a reference matrix followed by four information carrying matrices; however, the receiver uses an observation interval of $T = 4M$ with $\alpha = 1/(k-1)$, $\beta = 0$ as in (25), as assuming larger T was inefficient in SDE schemes.

Table I also compares the performance of the extended observation interval *overlap* scheme SDE, $k = 4$ to the basic *overlap* scheme denoted in the table as SDE $k = 2$. The channel in this case was realistically characterized for simulation by the correlation function in (2).

It may be noticed in the *overlap* schemes that SDE, $k = 4$ compared with SDE, $k = 2$ has a performance gain of 0.6 dB. Similarly, for the *non-overlap* schemes, the RDE $k = 5$ compared with RDE/SDE, $k = 2$ has a performance gain of 0.56 dB, when the inherent gain, which results from reducing the proportion of the reference symbols to the information symbols, in the *non-overlap* case, is first subtracted (e.g., for RDE, $k = 5$ performance gain $= 5.8 - 10\log_{10} 0.8/0.5 - 3.2$). It may be observed that in the *non-overlap* scheme, the RDE, $k = 5$ performs slightly better than the SDE, $k = 5$ proposed system. This was also evident in the single antenna case [34].

Although it is not simulated here, we expect the SDE to be more robust than the RDE in the *overlap* scheme when $k > 2$ since *only* in SDE can the receiver employ a sliding observation window, shifting the observation interval for every codeword $\underline{C}$.

D. Performance Comparison With Other Systems

In [12, Fig. 5], the simulation performance of a turbo code concatenated with a unitary modulation code is given for two transmit and two receive antennas, with $T = 5$, $N_{\rm inf} = 3500$, $R_s = 1.75$, and turbo code rate of $R_c = (7/8)$. A BER of 10^{-4} is achieved there at SNR $= 16$ dB. We simulated a system in the *non-overlap* scheme where $T = 4$, $N_{\rm inf} = 3000$, $R_s = 1.6875$

with 8-PSK symbols constructing the 2×2 unitary matrices D_i, which implies 9 bits for each codeword. The turbo-code rate is $R_c = (3/4)$ with the same generating polynomials. We achieve a BER of 10^{-4} at SNR $= 13$ dB, which is a 3-dB improvement in SNR. This result demonstrates the significant contribution of the feedback in the iterative joint demodulation and turbo-decoding process. Furthermore, it is demonstrated in Fig. 6 and in [12] that performance improves by 2–3 dB when applying iterative joint detection.

E. Non-Overlap versus Overlap

Significant advantage is achieved by the use of overlapping when the channel fading rate is relatively slow. We compare here the example of the *non-overlap* scheme with the *overlap* one. In the *non-overlap* case, $T = 4$, $N_{\text{inf}} = 3000$, and $R_s = 1.6875$, with 8-PSK symbols constructing the 2×2 unitary matrices D_i, turbo-code rate is $R_c = (3/4)$, and a BER of 10^{-4} is achieved at SNR $= 13$ dB. In the *overlap* scheme, $T = 4$, $N_{\text{inf}} = 3000$, $\tau = 64$, a higher channel throughput of $R_s = 2.25$ with 4-PSK symbols constructing the 2×2 unitary matrices D_i, and turbo-code rate of $R_c = (3/4)$, a BER of 10^{-4} is achieved at SNR $= 9.9$ dB, which is a 3.1 dB gain in SNR, sustaining an increased channel throughput by a factor of 1.33. Most of this improvement may be attributed to the overlapping of the observation intervals at the receiver, which is inherent to the *overlap* scheme.

VI. Conclusions

We demonstrated by simulations that turbo codes concatenated with differential space-time modulation codes can achieve a BER that is lower than 10^{-4} at 0.8 to 2.8 dB from a code-restricted mutual information limit for coding rates of 0.5 to 2.25 bits/channel use over the flat fading channel in two fading scenarios: *overlap* and *non-overlap*, respectively. Unitary matrix differential modulation was used to exploit channel characteristics. Iterative turbo decoding and joint demodulation performs closely to capacity limit and exhibits significant improvement over a more traditional decoder without the side information feedback. This improvement is even more pronounced over short block lengths.

From the comparison in Section V-D of our work to some results in [12], it may be seen that a significant improvement in SNR is gained. This emphasizes the contribution of processing the modulation code inside the iteration loop.

Optimization over the turbo component code selection and its contribution to performance and use of nonsquare information matrices are subject for future study.

As expected, overlapping inherent in differential schemes yields a significant performance advantage when applicable. The *non-overlap* scheme is directly applicable to fast frequency hopping spread spectrum communication, and the *overlap* scheme is applicable to a mobile-wireless channel when fading dynamics are relatively slow.

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